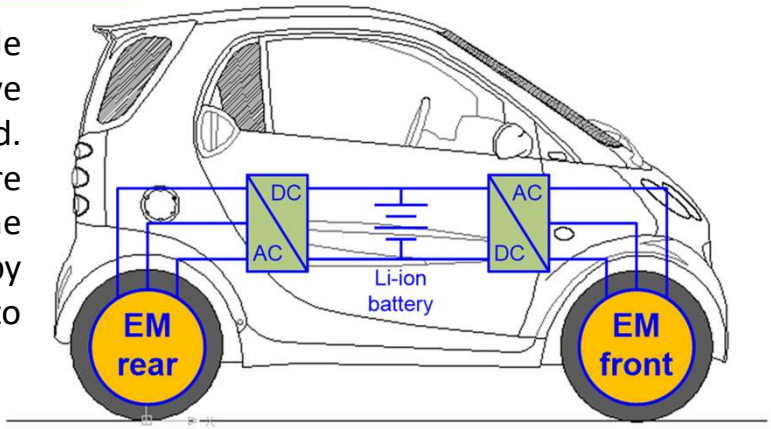
IEEE VTS Motor Vehicles Challenge 2021

ENERGY MANAGEMENT OF A DUAL-MOTOR ALL-WHEEL DRIVE ELECTRIC VEHICLE

In this fifth IEEE VTS Motor Vehicle Challenge, a dual-motor all-wheel drive (AWD) electric vehicle (EV) is studied. Two different electrical motors (EMs) are installed in the front and rear axes of the EV, respectively. The EMs are driven by the two inverters connected in parallel to a Li-ion battery.



THE CHALLENGE

The challenge is to develop an Energy Management Strategy (EMS) to reduce the motor drives losses in order to extend the range of the vehicle. The MATLAB/Simulink model of the studied AWD EV will be provided. Participated teams will develop their own EMSs to be compared on this common model. There will be two samples of speed profiles provided to the teams to test their EMSs. The submitted EMSs will be evaluated by using a third unknown profile. The participants will be invited to attend the IEEE VPPC 2021.

PRIZES

First prize: **US$3500** grant to attend the IEEE VPPC 2021
Second prize: **US$1500** grant to attend the IEEE VPPC 2021
This competition is open to everyone (students, academics, and industry). However, *only VTS members* are eligible to receive the grant, so JOIN US NOW!

COMMITTEE

Challenge Committee Chairs
João Pedro F. Trovão, Université de Sherbrooke, Canada
Samir Jemeï, Université de Franche-Comté, France
Loïc Boulon, Université du Québec à Trois-Rivières, Canada

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Source: e-TESC Lab and Association des Transports Électriques de l'Université de Sherbrooke

IMPORTANT DATE

Registration: December 20, 2020
Submission: February 22, 2021
Results: March 15, 2021

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http://www.motorvehicleschallenge.org/